Teleoperation

The Holy Grail to Solve Problems of Automated Driving? Sure, but Latency Matters

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ABSTRACT

In the domain of automated driving, numerous (technological) problems were solved in recent years, but still many limitations are around that could eventually prevent the deployment of automated driving systems (ADS) beyond SAE level 3. A remote operating fallback authority might be a promising solution. In order for teleoperation to function reliably and universal, it will make use of existing infrastructure, such as cellular networks. Unfortunately, cellular networks might suffer from variable performance. In this work, we investigate the effects of latency on task performance and perceived workload for different driving scenarios. Results from a simulator study (N=28) suggest that latency has negative influence on driving performance and subjective factors and led to a decreased confidence in Teleoperated Driving during the study. A latency of about 300 ms already led to a deteriorated driving performance, whereas variable latency did not consequently deteriorate driving performance.

CCS CONCEPTS

• **Human-centered computing → Human computer interaction (HCI)**; • **Networks**; • **Computer systems organization**;

KEYWORDS

latency, qos, teleoperated driving, user study

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AutomotiveUI '19, September 21–25, 2019, Utrecht, Netherlands
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ACM ISBN 978-1-4503-6884-1/19/09…$15.00
https://doi.org/10.1145/3342197.3344534

ACM Reference Format:
Stefan Neumeier, Philipp Wintersberger, Anna-Katharina Frison, Armin Becher, Christian Facchi, and Andreas Riener. 2019. Teleoperation: The Holy Grail to Solve Problems of Automated Driving? Sure, but Latency Matters. In *11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '19), September 21–25, 2019, Utrecht, Netherlands.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3342197.3344534

1 INTRODUCTION

Technological progress usually aims to change society to the better, and the list of potential advantages of automated driving technology is long. Automated vehicles (AVs) promise to reduce driver stress, parking costs, energy consumption, pollution, while increasing safety, productivity, mobility for non-drivers, or road capacity [21]. However, when looking on the streets today, we might realize that many of these advantages are for the long haul. Considering the SAE levels of automation [30], existing automated driving systems operate on level 2, and fully automated level 5 vehicles are not expected within the next years – actually, even when the technology is reliable, additional time will be needed for testing and regulatory approval [21]. In addition, recent incidents with AVs raised the question, if automation that requires a human driver as fallback authority can safely be implemented [11, 33]. A promising approach to solve such problems and bring AV technology to the customer earlier is Teleoperation. Teleoperated Driving (sometimes also referred to as "remote driving") is the remote control of a vehicle by a human in particular situations. For example, vendors of shared mobility could provide (level 3) vehicles that, for passengers, appear like fully automated vehicles, yet an off-board operator takes over if necessary. Another example are automated trucks, that may be able to drive alone on highways, but need support when reaching their operational design domain – in such scenarios, a small number of operators could potentially administer a larger number of vehicles. Further scenarios are software/hardware failures

of highly autonomous vehicles [17]. Teleoperated Driving systems are already in use and are being developed further by different companies [6]. It is additionally required for example in California to allow driverless vehicle testing [7]. Thus, Teleoperated Driving is claimed to play an important role in the story of autonomous vehicles. Enabling Teleoperated Driving requires the utilization of wireless connections between the operator and the vehicle [27]. A technology that is already widely deployed has to be used to allow operation over a broad geographic area, and cellular networks could become the technology of choice, since they are able to meet required latency, bandwidth and packet loss demands [18]. Unfortunately, remote control will always go hand in hand with the problem of latency, emerging from the need to send/receive vehicle states and control signals from and to vehicles on the road. This paper tries to answers the question, how latency influences average human drivers in safely controlling a remote vehicle. To figure out the impact of latency, we conducted a driving simulator study, where participants had to drive different simulated tracks and deal with various levels of potentially changing latency. Initial investigations in this regard have already been published (e.g. [22]), yet a fine-grained evaluation concentrating on quantitative evaluation of driving performance measures has not been made.

2 RELATED WORK

A detailed investigation of about 150 papers regarding different aspects of human performance issues and mitigations for remote operations was done by Chen et al. [3], with the aim to point at areas where further research is required. Lichiardopol did a survey on teleoperation [20], presenting a general overview of teleoperation and covering important aspects such as telepresence and control issues. The basic concept of teleoperated robotics has been shown by Winfield in [35]. There, a teleoperated system consists of the three essential parts; remote device, teleoperation workspace and connection between both. Basic setups for Teleoperated Driving are shown by Neumeier et al. [27], Gnatzig et al. [13] and Shen et al. [32].

The ability to provide immersion, the telepresence, is a relevant prerequisite for Teleoperated Driving [2] and widely investigated. In [16] Hosseini and Lienkamp show that Teleoperated Driving could be improved by virtual-reality systems. Virtual-reality systems and multi-monitor setups are also compared by Georg et al. [12]. It is shown that head-mounted displays allow for a better immersion, but - in contrast to the aforementioned results - do not necessarily improve driving performance. Chucholowski [5] analyzed different display methods for teleoperated vehicles and showed that a predictive display can help to mitigate time delays. In [16] Hosseini and Lienkamp introduce a combined approach with virtual-reality and augmented sensor data to make driving safer.

To use Teleoperated Driving in a safe manner, the network is one of the most important factors. Network latency for Teleoperated Driving has been measured by Shen et al. [32], showing a average latency for video streaming of about 183 ms and 205 ms and an average latency of about 110 ms and 217 ms for vehicle controls in 4G and 3Gb respectively. It has been shown that driving a slalom course was possible even with 3G. Chucholowski et al. [4] faced an average latency of about 121 ms when submitting a video stream through 3G networks. Real-world measurements of cellular networks have been done by Neumeier et al. [28], where a median of Round-Trip-Time (RTT) of about 55 ms for UDP- and Ping-based measurements was seen. Kang et al. [17] set up a real-world testbed to analyze the behaviour of LTE and WiFi for Teleoperated Driving, showing that the median two-way latency was about 100 ms over the LTE network. One of the major considerations in Teleoperated Driving is the present existent latency, caused by the use of telecommunication networks. Luck et al. [24] investigated the effects of latency on remote operation and claim that constant latency is easier to deal with than variable one. This claim is confirmed by findings of Davies et al. [8] and Liu et al. [22].

User studies regarding Teleoperated Driving have been carried out by various research groups. In [22] Liu et al. conducted an user study with state-of-the-art LTE network performance and a small-scale vehicle. They claim that Teleoperated Driving over LTE does not work without supporting systems. Vozar and Tilbury [34] conducted an user study to explore the effects of latency. It is shown that the path-following score decreases with higher latency. A further, not Teleoperated Driving specific, user study was conducted by Nielsen et al. [29]. They introduced a combined 3D view and analyzed the results, showing that their approach improves the driving.

Although there is already a lot of research on Teleoperated Driving, including user-studies, most of it consists of single, small-scale scenarios. With the user study conducted in this paper, we overcome these issues by using more realistic, close to real-world scenarios combined with real-world measured latencies.

3 USER STUDY

We conducted an user study to identify the impact of latency on driving performance, workload, and user acceptance. The study took place in a static driving simulator with a multi-monitor setup as used by "Designated Driver" [7] and "Phantom Auto" [14]. Consequently, motion sickness potentially caused by artificial latency should be limited. We implemented different scenarios, along with different fixed and variable latencies, aiming to get close to real-world driving situations. The chosen scenarios reflect real-world driving, including parking and general vehicle handling. By

evaluating both, behavioral (driving performance) and self-rating (e.g. NASA-TLX [15], self-defined questionnaire) data, we thereby want to answer the following research questions:

- **RQ1:** How do different constant latencies influence driving performance?
- **RQ2:** Does constant latency lead to better driving performance than variable latency?
- **RQ3:** What impact does latency have on subjective workload?

Setup and Scenarios

The experiment was conducted with an internally developed 3D driving simulator called *OpenROUTS3D* [25]. It allows for the creation of dynamic scenarios with different levels of input/output and inter-frame latency by applying buffers. Latency was configured for input and output to simulate both the latency of control commands, and the latency of a video stream, respectively. Latency was not displayed with its real value, but within groups indicated by colors and text. Groups were: no (0 ms), low (1 - 300 ms), medium (301 > 500 ms) and high (> 500 ms) latency, to match real-world Teleopearted Driving, where remote operators usually at least know the current level of latency to react appropriately. The physical driving simulator setup can be seen in Figure 1. Participants sat in the driving simulator with three 32" low-latency curved monitors (*AOC C32G1* with 1 ms Moving Picture Response Time (MPRT) [1]) arranged in a horizontal row. The chair was a *Playseat Evolution Alcantara* [31] gaming racing chair with *Logitech G29* steering and pedals [23] mounted to it. The driving simulator was separated from the investigators by movable partition walls. Thus, the participant was in his own private area, not observable by others during the study. However, the participant was able to talk to the investigators at any time if required. We abstained from using virtual reality head-mounted displays to avoid motion sickness during the study, which had a duration of up to 1 hour. Such a system could improve telepresence [16], but it does not necessarily lead to better driving performance, as more participant training with virtual reality systems might be required [12].

Considering latency as the main influencing factor, the driving simulators's baseline end-to-end latency had to be known. Thus, the delay between steering wheel command and corresponding display output was measured. In order to do so, one of the steering wheel buttons was used to switch a rectangle on the center monitor from black to white. Using an external microcontroller with attached push button and photodiode allowed precise measurements of the time delays between button pressed and display output. A TSL250R photodiode was mounted on the front monitor within the



Figure 1: Setup of the driving simulator study.

color-changing rectangular. According to the datasheet, the sensor has an output rise-time of about 260 microseconds. As soon as the display output changes from black to white, a hardware interrupt on the microcontroller is triggered. The measurements revealed a median base latency (button pushed to display change) of about 66 ms, based on 50 measurements. It was also checked if the artificial input and output latency was applied correctly. Setting an artificial latency of 500 ms (250 ms each for input and output), the mean end-to-end latency was 567 ms, based on 21 measurements. This result was expected from the first measurements and thus latency was determined to be correctly applied. The framerate was set to constant 60 FPS. A more detailed explanation of the measurement setup in addition with the Unity-based source-code can be found as OpenSource-project on gitlab[1].

The user study consisted of five different scenarios. Every map was driven during daytime with sunny weather and did not contain any visible elements that are not driving-related. The streets had a width of about 7.0 m and thus followed the German "Richtlinien für die Anlage von Landstraßen (RAL)" [10]. The first map every participant drove was the practice map (see Figure 2). The green line indicates the center line of the track, whereas the red dot was the starting position of the drive. This map was driven without latency and ended either by user action (pressing button) or reaching the time-limit of 200 seconds. The scenarios for the experimental trials were `Parking`, `Snake`, `Pylon` and `LongTrack`. The most trivial scenario was the `Snake`, shown in second place in Figure 2. On this map the user had to stay on the street and follow it until reaching the final markers. This map contained some curves and was intended to check if it is possible to follow a simple street without complex steering maneuvers.

The `Pylon` map (third in Figure 2) consists of one big double curve with pylons placed on the center line of the track.

[1] https://gitlab.com/becheran/button-to-photon-latency-measure/

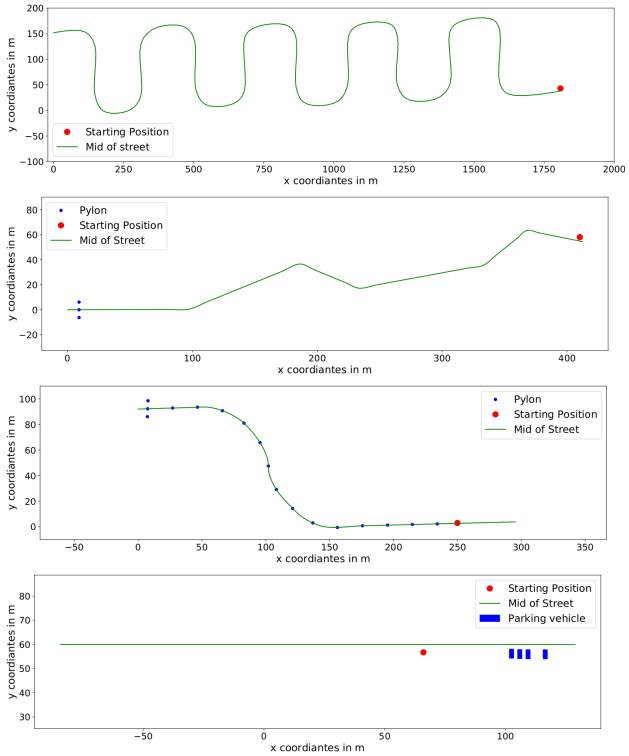


Figure 2: Maps of the user-study, including LongTrack/Practice, Snake, Pylon and Parking (from top).

Participants were instructed to drive in a zigzag pattern between the pylons. The curve was added to avoid acclimation of the driver as if he had only been driving on a straight street. This map was used to evaluate how participants behave in a handling course.

The Parking map (fourth in Figure 2) consists of four vehicles and a parking lot. Participants were instructed to park the vehicle inside this simulated parking lot and were allowed to decide if they preferred forward or backward parking (only with mirrors). The purpose of this map was to see if it is possible to safely execute a trivial everyday driving task, such as parking.

Finally, participants were required to drive the LongTrack scenario, which is also the first map in Figure 2. This track is the same as were introduced with the practice map, but with the difference that latency changed whilst driving was underway. The different areas, indicated with different colors, can be seen in Figure 3. Participants were instructed to follow the street, which was chosen to simulate a curvy rural road. The latency was changed to see how changing latency influences driving performance. This could happen frequently in real-world driving scenarios.

Every map (except the LongTrack map) was driven with three different levels of latency. The latency was either 0.0 s

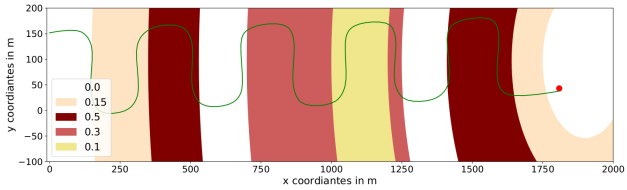


Figure 3: Latency areas for the LongTrack scenario.

(**no**), 0.15 s (**small**) or 0.3 s (**large**) each for input and output latency. The LongTrack had **varying** area-dependent delays between 0.0 s, 0.05 s, 0.075 s, 0.15 s and 0.25 s each for input and output latency. These delays were chosen based on the findings of Davis et al. [8] and Kang et al. [17], where a maximum acceptable latency of 700 ms [8] was identified. They also match with the results of real-world measurements conducted by Neumeier et al. [27]. Results of latency in video games as shown by Dick et al. [9] were also considered. Thus, latencies were selected to cover the range of previous findings, considering the constant presence of about 60 ms system latency.

Measurements

For a holistic contemplation of participants' teleoperated driving experience, we triangulated objective driving performance measures with participants' subjective self-ratings, which we collected by a questionnaire.

Driving Performance. To record driving performance, the current position of the pedals and steering wheel, the position of the vehicle and the current latency were logged along with additional information every frame (approximately every 16 ms). This log was analyzed with respect to the research questions and thus the following metrics were taken into account:

- **Lateral Deviation (MLP, SDLP)**: For position on the street, the Mean Lateral Position (MLP) and Standard Deviation of Lateral Position (SDLP) have been calculated [19] based on the center line. This values helped to indicate if driving was accurate or if there was a lot of deviation in the driving task. For the calculations, the vehicle's center point was used.
- **Max. Steering Angle**: This parameter describes the maximum angle participants steered to fulfill the driving task. This value (maximum of 35° in steering) helped to indicate how smooth a ride was and how participants were able to control the vehicle. The lower (except the Parking scenario) the value, the smoother a ride was (under the assumption that a large deviation from the track was compensated with sharp steering maneuvers).

- **Out of Lane Ratio**: This value represents the time (relative to the whole duration of a trial), in which a vehicle was not on the street. It was counted if at least one wheel crossed a side-lane marking. This value contributed to an understanding of the driving ability of the participant.
- **Avg. Speed**: Average speed indicated how fast participants were able to drive during a task.
- **Acceleration/Deceleration**: These values were calculated based on the speed-differences taken over time and only considered actions intended by the user. Thus, only accelerations and decelerations, which changed the vehicle's absolute speed at least (+/-) 2 km/h, were considered. They were assumed to be intended actions by the participants. Otherwise all rolling and speed maintenance actions would have been considered, but were not of interest to this evaluation. Both values could be used to determine the smoothness a drive had. The stronger the deceleration and acceleration, the bumpier the ride, thus, lower values are (potentially) better than higher ones.

Questionnaire. The questionnaire consisted of demographic information about age, gender, driving and video gaming experience to obtain some basic information on the participants (pre-driving questions). In addition, we assessed participants' trust in automated driving and public transport (*"I trust an automated vehicle/ public bus controlled by a computer/ by a teleoperated driver."*), as well as their confidence to remotely control a vehicle with, and without passengers on their own (*"I feel confident to be able to drive a vehicle with/without passengers teleoperated."*). To analyze the difference between expectations and reflection, we asked these questions before and after the drives (pre- and post-driving questions). To analyse the impact of the different latencies on participants' experience (post-latency questions), we utilized the NASA Task Load Index (NASA-TLX) [15] to quantify cognitive workload. Further, we implemented a self-defined questionnaire (in German, using a 5-point Likert scale). Thereby, we aimed to assess participants' stress (*"I perceived the situation during the trip as very stressful.", "I felt overwhelmed during the last ride.", Cronbachs* $\alpha > 0.8$), driving comfort (*"I perceived the situation during the trip as very comfortable."*), controllability (*"I had the vehicle well under control."*), and appropriateness of remote control of the experienced latency (*"I find the situation I have just experienced suitable for road traffic."*). Moreover, participants' were asked about the reasons of their problems regarding a specific latency.

Procedure

We utilized a within-subjects experimental design, where each participant completed each scenario multiple times, and with all combinations of scenario and latency, leading to a total of 10 trials (the three scenarios `Snake`, `Parking`, and `Pylon` with **no**, **small**, and **large** latency, as well as the `LongTrack` with **varying** latency) after the practice run. The order of conditions was randomized on two levels, each participant experienced a randomized order of latencies and within each latency, a randomized order of scenarios. At the beginning of the study, participants were introduced to the simulator environment and the basic concept of Teleoperated Driving. After answering the pre-driving questions, participants practiced driving on a test track for a maximum of 200 seconds.

Subsequently, the first of four relevant driving blocks started. A block consisted either of one fixed latency (**no**, **small** or **large**) with all maps (`Snake` and `Pylon` and `Parking`) except `LongTrack` in random order or the **variable** latency-based `LongTrack`. Thus, every participant drove the `LongTrack` once (if not considering practicing) and all other tracks three-times, once for each latency. The driving goals of each combination of map and latency were shown prior to every drive, e.g. follow the street, drive through the pylons, park between the cars. After each block, participants had to answer the post-latency questions. Subsequent to the final block, the questionnaire contained the additional post-driving questions. The procedure for each participant can be seen in Figure 4.

Participants

In total, 28 participants (6 female) took part in the driver study (M_{age}=26.89, SD=7.72). As we assume future teleoperated drivers to be young and tech-savvy, who might be, but not necessarily, interested in video gaming, we choose a diverse sample in the age between 10 and 30. All participants had a driver's license. Most of them drive more than 1,000 km per year (about 90%), while 2 participants drive more than 25,000 km per year. Self-reported driven distances ranges from 25% of participants with less than 5,000 km per year, 17% between 5,000 km and 10,000 km per year, 17% between 10,000 km and 15,000 km per year, 32% between 15,000 km and 25,000 km per year and 7% above 25,000 km per year. Participants' video game experience is diverse. It ranges from 35% who never play, to 40% who play once in a while, to only 25% who play frequently. Most played games are Shooter games (30%), followed by Strategy (26%), Jump and Run (12%), Simulators, Racing and Puzzle games (each 7%). Least played games are MMO and party games (each 5%).

4 RESULTS

Results are based on the different levels of latency and scenario, which accounts for the order throughout this section.

	Pylon Mdn (IQR)			Snake Mdn (IQR)			Parking Mdn (IQR)			Long Track Mdn (IQR)
Latency	**no**	**small**	**large**	**no**	**small**	**large**	**no**	**small**	**large**	**varying**
Lane Keeping Performance										
MLP	1.85 (.83)	2.13 (.55)	2.21 (.87)	2.52 (.73)	2.73 (.89)	2.66 (.52)	-	-	-	2.94 (.54)
SDLP	1.07 (.40)	1.27 (.45)	1.41 (.54)	.96 (.36)	1.07 (.58)	1.30 (1.02)	-	-	-	1.59 (.78)
Max. Steer Ang.	23.80 (8.10)	27.95 (8.48)	35.00 (6.07)	13.30 (4.55)	14.15 (5.90)	16.75 (13.92)	35.00 (6.2)	35.00 (5.65)	35.00 (8.10)	15.55 (7.48)
Out of Lane Ratio	0 (0)	0 (.03)	.03 (.08)	0 (.01)	0 (.03)	.02 (.08)	-	-	-	.027 (.05)
Speed and Acceleration/Deceleration										
Avg. Speed km/h	20.07 (5.49)	17.93 (8.60)	14.94 (6.18)	38.31 (13.92)	34.56 (13.38)	31.39 (9.66)	4.29 (3.31)	4.57 (4.46)	4.00 (3.03)	53.28 (3.31)
Mdn. Acc m/s^2	1.13 (.97)	1.15 (.97)	1.28 (.72)	1.23 (.62)	.96 (.90)	1.62 (1.11)	.80 (.36)	.85 (.66)	.89 (.40)	1.75 (.50)
Mdn. Decc m/s^2	3.05 (4.30)	2.84 (3.95)	2.14 (4.27)	1.53 (4.28)	5.82 (3.77)	4.38 (2.63)	1.36 (1.03)	1.76 (1.33)	1.64 (1.30)	4.99 (1.18)

Table 1: Medians and interquartile ranges of collected driving performance measures, showing the negative influence of increasing latency.

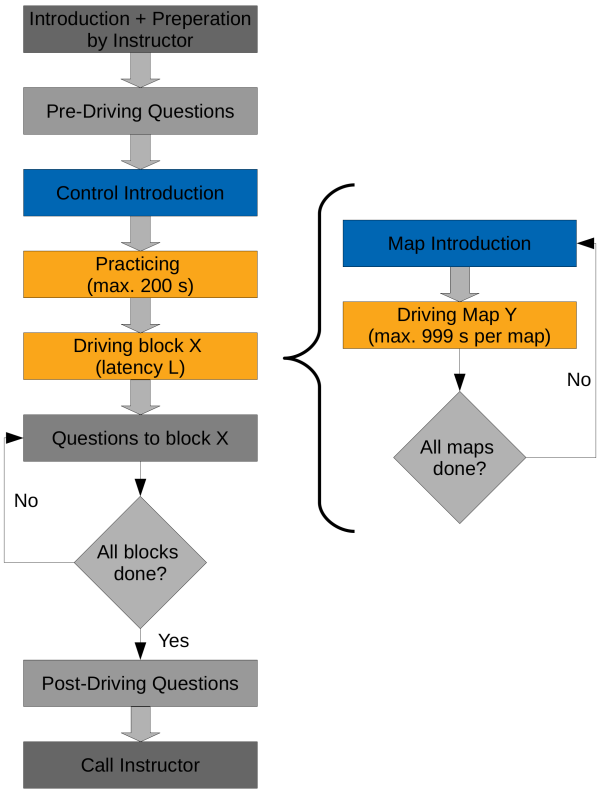


Figure 4: Procedure of the user study.

Driving Performance

To compare driving performance between the different latencies, we conducted multiple Friedman ANOVAs (data not normally distributed) within the given scenarios, since a direct comparison between the individual scenarios would not make much sense due to inherently different road configurations. However, to account for multiple tests given the 9 individual conditions, we adjusted the significance level accordingly to $.05/9 = .00\overline{5}$. In the following, we report the results of the statistical evaluation. An overview of descriptive statistics is depicted in Table 1.

Pylon Scenario. Considering the Mean Lateral Position (MLP), Friedman ANOVA indicated a significant difference between the three latencies ($\chi^2(2) = 12.07, p = .002$), whereas pairwise comparisons reveal differences between **no** and **small** ($p = .01$), and **no** and **large** latency ($p = .006$). However, both did not satisfy the alpha adjusted significance level. The Standard Deviation of Lane Position (SDLP) shows a similar picture, here Friedman ANOVA highlighted a significant effect too $\chi^2(2) = 11.14, p = .004$, and while the difference between **no** and **small** latency again was dismissed due to alpha correction ($p = .048$), we can accept the effect for the difference between **no** and **large** latency ($p = .004$). The Maximum Steering Angle for the `Pylon` scenario ($\chi^2(2) = 27.42, p < .001$), on the other hand, indicates significant differences between **no** and **small** ($p = .002$), as

well as **no** and **large** latency ($p < .001$). Similar for the Out of Lane Ratio ($\chi^2(2) = 19.70, p < .001$) – in the **large** latency condition, participants got out of the lane significantly more often, than in the lower latency conditions ($p = .001$).
Regarding driving speed and forces resulting from gas/brake pedal actuation, significant differences were found for the average speed ($\chi^2(2) = 30.50, p < .001$) between **no** and **small** ($p < .001$) and **no** and **large** ($p < .001$) latency, but no effects could be obtained for acceleration/deceleration values.

Snake Scenario. In the Snake scenario, no differences could be obtained for MLP, but Friedman ANOVA indicated a significant effect for SDLP ($\chi^2(2) = 13.07, p = .001$), that was significantly higher in **large**, compared to **no** latency ($p = .001$). Similar, the Maximum Steering Angle ($\chi^2(2) = 12.07, p = .002$) was significantly higher in the **large** than in the **no** latency condition ($p = .002$). No differences could be revealed for the Out of Lane Ratio.
Still, in this scenario various effects for speed and gas/brake pedal actuation were present. Considering speed ($\chi^2(2) = 33.50, p < .001$), significant differences were found between **no** and **large** ($p = .002$), as well as **small** and **large** ($p = .002$) latency. Also for acceleration ($\chi^2(2) = 12.50, p = .002$), a difference was present between **no** and **large** ($p = .003$) latency, but not for other pairwise comparisons. In contrast for deceleration, again both **small** ($p = .001$), and **large** ($p = .005$) latency differed significantly from the **no** latency condition.

Parking Scenario. Interestingly, in the `Parking` scenario no significant differences could be found. Here, we did not calculate lane keeping performance parameters (it was a very short scenario, where leaving the lane to drive into the parking spot was given by definition, resulting in the maximum steering angle in all cases), but also regarding speed, acceleration, and deceleration, no effects are present.

LongTrack Scenario. To quantitatively compare **varying** and fixed latency, we choose to evaluate only the Lane Keeping parameters MLP and SDLP in comparison to the `Snake` scenario. Other parameters would certainly differ due to the differences in the road configuration, and also MLP and SDLP must be considered with caution, but we still claim (especially SDLP) as a valid measure for comparison, since it is typically chosen to evaluated driving errors. For statistical evaluation we conducted a series of pairwise comparisons using Wilcoxon Signed Rank tests suitable for within-subjects evaluation.
Considering MLP, all investigated latencies **no** ($Z = -3.67 p < .001$), **small** ($Z = -2.48 p = .013$), and **large** ($Z = -2.66 p = .008$) were significantly smaller than the values obtained in the `LongTrack`, assuming an alpha-corrected significance level of ($.05/3 = .01\overline{6}$) to account for multiple comparisons. In contrast, the SDLP differed only significantly to the **no** latency condition ($Z = -4.01 p < .001$) – the **small** latency indicated a significant difference ($Z = -2.28 p = .023$), that however did not match the corrected alpha level.

	Pre TD Mdn (IQR)	**Post TD Mdn (IQR)**
With passengers	2.00 (2.00)	0.50 (1.00)
Without passengers	3.00 (2.00)	2.00 (2.00)

Table 2: Medians and interquartile ranges of participants confidence to operate an vehicle.

Questionnaire

Pre- vs. Post-Driving. Regarding general questions before and after the experiment, a Wilcoxon signed-rank test (due to the nature of Likert scales) revealed that participants' trust in teleoperated vehicles (z=-2.99, p=.003) and public transport (z=-3.47, p = .001) significantly decreased after the experience of Teleoperated Driving (for both before: Mdn=2.0; for both after: Mdn=1.0). Further, participants' confidence to operate a teleoperated vehicle with (z=-4.11, p< .001) and without passengers (z=-3.12, p=.002) on their own significantly decreased as well. Thereby, we could identify a significant difference of feeling confident to operate a vehicle with or without passengers, when they were asked after experiencing Teleoperated Driving (z=3.90, p<.001), see Table 2.

Differences between Latencies (Self-Defined Questions). To compare the different latencies, we applied Friedman ANOVAs with Bonferroni correction for multiple comparison. Participants' overall stress significantly differs between all conditions ($\chi^2(3) = 26.83, p < .001$). Pairwise comparisons revealed that stress differentiates significantly between **no** and **large** (p <.001), between **no** and **varying** (p=.003), and between **small** and **large** latency (p<.023). Participants perceived comfort differs also between all conditions ($\chi^2(3) = 45.64, p < .001$). Pairwise comparisons showed a significant difference between **no** and **small** (p=.027), between **no** and **large** (p<.001), between **no** and **varying** (p<.001), and between **small** and **large** latency (p=.007). Perceived control differs also significantly ($\chi^2(3) = 47.03, p < .001$), pairwise comparison reveal differences between **no** and **small** (p=.016), **no** and **large** (p<.001), **no** and **varying** (p<.001), **small** and **large** latency (p=0.008). Thereby also the rated appropriateness of Teleoperated Driving on real road is significantly affected by different conditions ($\chi^2(3) = 33.86, p < .001$). However, a significant effect could only be identified

	No Mdn (IQR)	Small Mdn (IQR)	Large Mdn (IQR)	Varying Mdn (IQR)
Stress	0.50 (1.00)	1.00 (2.38)	2.75 (2.38)	2.00 (1.50)
Comfort	3.00 (2.00)	2.00 (2.00)	1.00 (2.00)	1.00 (2.00)
Control	3.00 (1.00)	2.00 (2.00)	1.00 (2.00)	1.50 (3.00)
Appropriateness	2.50 (1.00)	2.00 (1.00)	1.00 (1.00)	1.00 (2.00)

Table 3: Medians and interquartile ranges from the post-latency questions.

	No Mdn (IQR)	Small Mdn (IQR)	Large Mdn (IQR)	Varying Mdn (IQR)
Effort	1.00 (1.00)	2.00 (2.00)	3.00 (1.00)	2.00 (1.00)
Frustration	1.00 (1.00)	1.00 (3.00)	3.00 (1.00)	2.00 (3.00)
Performance	1.00 (2.00)	2.00 (2.00)	3.00 (3.00)	2.00 (1.00)
Mental Demand	1.00 (2.00)	1.00 (2.00)	3.00 (2.00)	3.00 (1.00)
Physical Demand	0.00 (1.00)	1.00 (2.00)	2.00 (3.00)	1.00 (2.00)
Temporal Demand	0.50 (1.00)	1.00 (2.00)	1.00 (2.00)	1.00 (2.00)
Overall Workload	4.50 (5.75)	7.50 (6.00)	13.00 (8.50)	10.00 (5.75)

Table 4: Medians and interquartile ranges from the Nasa-TLX questionnaire.

between **no** and **large** (p<.001), and **no** and **varying** latency (p=.001), see Table 3.

Differences between Latencies (NASA-TLX). Regarding participants' workload, results from the NASA-TLX (Table 4) reveals significant differences for the overall perceived workload (i.e., sum of all sub-scales), ($\chi^2(3) = 52.42, p < .001$). Thereby, pairwise comparisons reveal significant effect between **no** and **small** (p=.023), **no** and **large** (p<.001), **no** and **varying** latency (p<.001), and **small** and **large** latency (p<.001). Regarding sub-scales individually, significant effect could be revealed in all sub-scales: effort (E: $\chi^2(3) = 27.41, p < .001$), frustration (F: $\chi^2(3) = 31.17, p < .001$), performance (P: $\chi^2(3) = 30.71, p < .001$), mental demand (MD: $\chi^2(3) = 33.20, p < .001$), physical demand (PD: $\chi^2(3) = 25.88, p < .001$) and temporal demand (TD: $\chi^2(3) = 14.09, p = .003$). Thereby, pairwise comparison reveals significant differences between **no** and **large** (E: p<.001; F: p<.001; P<.001; MD: p<.001), and **no** and **varying** latency (E: p=.006; F: p=.010; P: p=.002, MD: p=.001) for the sub-scales effort, frustration, performance and mental demand. Participants' frustration significantly differs in addition between **small** and **large** (F: p=0.037), and **small** and **varying** latency (F: p=0.010), while mental workload differs additionally between **small** and **varying** latency (MD: p=.023). Physical demand only differs between **no** and **large** (PD: p<.001) latency. However, individual latencies do not affect the perceived temporal demand, no significant differences could be identified.

Problems of Teleoperated Driving. In the condition with **no** latency, 35.71% (n=10) of the participants reported to have no problems with Teleoperated Driving, 25.00% (n=7) complained about the clarity of the monitors, 14.29% (n=4) about the the control of the steering wheel, and 10.71% (n=3) about unpredictable delays during their input. Steering the pedals and lack of clarity of the driving route were only mentioned by respectively 7.14% (n=2).

Already with a **small** latency, only one participant reported to have no problems, however, 28.57% (n=8) complained about a constant delay during their input (not mentioned after no latency), 25% (n=7) complained about unpredictable delays, and respectively 17.86% (n=5) about controlling the steering wheel and clarity of the monitors. Only 7.14% (n=2) complained about the lack of clarity of the driving route.

After the condition with a **large** latency none of the participants reported no problems with Teleoperated Driving, or complained about the monitor clarity. However, here, constant delays (53.57%, n=15) and unpredictable delays (32.14%, n=9) were mentioned as major problems. Controlling the steering wheel (7.14%, n=2), pedals and lack of clarity of the route (respectively 3.57%, n=1) were mentioned only a few times.

While **varying** latency, 10.71% (n=3) reported to have no problems. The most mentioned problem is the unpredictability of the delays (60.71%, n=17). Controlling the steering wheel (10.71%, n=3), lack of clarity of the route and monitors (respectively 7.14%, n=2), and constant delays (3.57%, n=1) were mentioned only by a few participants.

5 DISCUSSION

In the following, we discuss the implications of our results on the proposed research questions. To get an overview on how a drive looked like, Figure 5 shows the latency-based results of a single participant on the `Pylon` track.

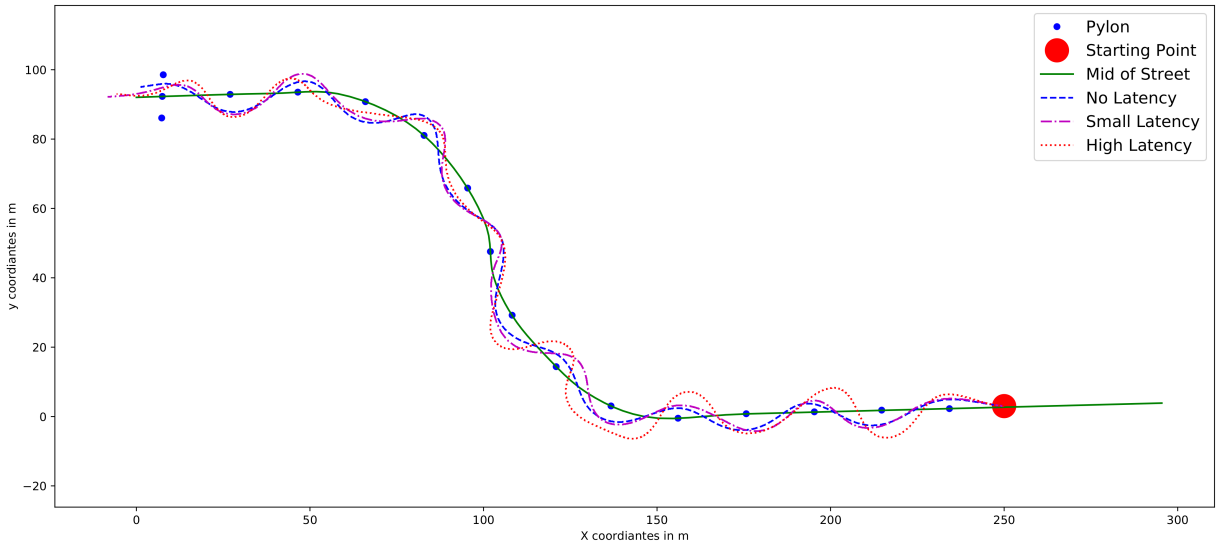


Figure 5: Driving results of one participant on the pylon scenario.

RQ1: Impact of Constant Latency on Driving Performance

The statistical evaluation of the different driving performance measures in the given scenarios reveals an interesting picture. While the **small** latency condition did not reveal a significant decrease for lane keeping (effects for MLP dismissed due to alpha-adjustment) performance, a significant difference could be obtained between **no** and **large** latency for SDLP in the Pylon and the Snake scenario. Since SDLP is a good predictor of lane keeping, this appears as driving with the **small** latency condition did not significantly worsen performance while **large** latency did. This might be supported by evaluation of the maximum steering wheel angle, that did not differ in the Snake, yet in the Pylon scenario between all investigated latencies. Also, participants significantly more often left the lane with **large** latency in the Pylon scenario (but no differences could be obtained for the **small** latency). In a similar fashion, speed and acceleration/deceleration differed only between **small** and **large**, as well as **no** and **large** latency in the Snake scenario. Overall consideration of these results makes us believe, that driving with **small** latency is acceptably good as with **no** latency, while **large** latency significantly worsens performance (comparable to the results of [34]). However, this has to be taken with caution - since due to the 9 conditions, the alpha-adjusted significance level was comparably low, and a higher number of participants might yield to additional significant effects. Still, we believe that the acceptable latency must be related to the actual driving scenario. In the Parking scenario, even no differences for whatever latency could be revealed, and most effects were visible in the Pylon scenario, that is somehow artificial compared to reality on roads (yet we believe it to be a good scenario for our investigations, and we see the high number of effects in this scenario as a confirmation). Summarizing, driving performance decreases with increasing latency, but in very slow scenarios higher latencies might be acceptable, while in complex scenarios that require many fast adjustments (as in the Pylon scenario), even **small** latency could be too much. In scenarios with low complexity (such as the Snake scenario), low latencies could be acceptable. However, with other traffic objects on the road, even here **small** latency could become dangerous, but on the other hand, low latencies below 300 ms round trip time might be sufficiently handled by trained operators (study participants engaged in remote driving for the very first time). Still, further experiments are needed in typical everyday situations with unpredictable outcome.

RQ2: Fixed vs. Varying Latency

When comparing lane keeping performance between the LongTrack (**varying** latency) and the Snake scenario (other comparisons would not be applicable due to the inherently different types of driving tasks), the mean lateral position was significantly worse on the LongTrack, compared to all latencies in the Snake scenario. However, also here the results could emerge from the different road configuration. A more realistic assessment could be provided by investigation of the standard deviation of lane position (SDLP). Here, differences were only visible to the **small** latency condition, but again the road configuration could potentially have influenced the results. We can thus (at least in our scenario comparison, given the defined latencies) not confirm the results of Kang

et al. [17] and others, who state that fixed latency leads to better driving performance than **varying**.

RQ3: Self-Rated Perception

Evaluation of subjective ratings for workload (NASA-TLX), as well as our self-defined items is only possible between the different latencies, as assessment was conducted only after experiencing all three different scenarios in a given latency. The overall workload of NASA-TLX is a good indicator to see how demanding different latencies were and if an average driver could deal with them. Results show, that significant differences in the overall workload are present between **no** and **small**, **no** and **large**, **no** and **varying** and **small** and **large** latency. For Teleoperated Driving this means, that besides the **large** and **varying** latency, even **small** latency leads to a significant difference in the overall workload and might already require trained operators. As there is a significance between **small** and **large** latency, the aim of every teleoperated system design should be a reduction of system latency, at least below the 300 ms, as trained operators might also struggle with higher latencies. This accompanies with the significant difference between **no** and **large** and **no** and **varying** latency, considering effort, frustration, performance and mental demand. Taking the self-defined questionnaire into account, participants felt significantly more stressed if driving with **large** or **varying** latency. The perception of comfort and control changed as latency was added artificially. A significant difference in the appropriateness of Teleoperated Driving in a specific scenario could be seen between **no** and **large** and **large** and **varying** latency, where **large** and **varying** latency are marked as inapplicable.
In general, the overall workload shows that there is already a difference between **no** and **small** latency. For applying Teleoperated Driving in real-world, a skilled and trained driver thus seems indispensable. This finding is assured by the multiselect answers participants gave, where latency (if present) was reported as a problem. From a system's perspective the latency should be kept constant and as low as possible, but at least below 300 ms.

6 LIMITATIONS AND FUTURE WORK

The study was carried out with a limited number of 28 participants. This can show a trend but is not sufficient to provide general conclusions. Participants were in a situation where no harm could be deployed by them or to them. Thus, it is likely that drivers would behave differently when driving vehicles in real-world and the possibility of harming others. Selected participants were not trained specific to Teleoperated Driving. Thus, they are not skilled in dealing with latency. Results with skilled and special-trained remote operators might differ. The user-interface was limited to the multi-monitor setup. Results might differ from the actual ones if utilizing a virtual-reality system setup [16]. The selected tracks and latencies only reflect an extract of real-world scenarios and did not consider other traffic participants. With latency the user-study only focused on one Quality of Service (QoS) aspect required for Teleoperated Driving. Besides the latency, bandwidth and packet loss are also important.

For future work the next step is to conduct a driver study with artificial traffic in real-world scenarios, e.g. driving through a city with artificial traffic and other QoS factors. The subsequent step is conducting real-world test drives. It will be further investigated if specially trained drivers are able to overcome latency and drive safely even with latency. This includes considerations if there are any specific attributes a remote operator should have. Finally, it has to be investigated which content is required for a remote driver and how this content can be displayed best, as some of the participants complained about varying latency even in scenarios where latency was constant.

7 CONCLUSION

We conducted an user study based on a virtual driving simulator to compare different levels of latency in Teleoperated Driving. The participants had to drive different typical real-world scenarios with either varying or constant latency. In addition to the objective driving performance, subjective workload and acceptance of teleoperation was assessed using questionnaires. Results show that latency has an influence on the driving performance. Even latency of about 300 ms led to a deteriorated driving performance. Variable latency did not consequently deteriorate driving performance. These objective results partly match with the subjective assessments of the participants, but large and variable latency lead to a higher overall workload. The confidence in Teleoperated Driving decreased during the user study as participants had to deal with constant/variable delay on their own. Thus it is important to consider all these results for real-world Teleoperated Driving. Mapping these results to real-world network measurements reveal, that today's cellular networks can provide required performance, but - as previously mentioned for example by [22] - further approaches (e.g. whitelisting [28], driver support systems and route planning [26]) are required. In general it can be said, that latency should be constant and as low as possible and only skilled and trained drivers should act as remote operators.

ACKNOWLEDGMENTS

This work is supported under the "Innovative Hochschule" program of the German Federal Ministry of Education and Research (BMBF) under Grant No. 03IHS109A (MenschIN-Bewegung), as well as the FH-Impuls program under Grant No. 13FH7I01IA (SAFIR). The authors would further like to thank Nino Bock for his support during the study.

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